

An AI research assistant for multi-omics discovery in marine microbes

Integrated published data → ask questions in plain language → reproducible analyses

Published multi-omics data

Genomes

Transcriptomes

Proteomes

Metabolomes

Annotations
& orthologs

1 KNOWLEDGE GRAPH

Prochlorococcus & Alteromonas — published multi-omics, integrated into one graph

Publication

studies

reports

Experiment

RNA-seq · proteomics

measures

Gene

central entity

in

Organism

strain / taxonomy

conserved

Ortholog group

cross-strain

annotated

Ontology

GO · KEGG · Pfam

acts on

Metabolite

reactions · transport

at a glance

120,000+

genes

43

published
studies

197

experiments

45

organisms

RESEARCHER

2 ASK QUESTIONS

In plain language or in Python — no query language to learn

**Expression
& DE**

**Function
& pathways**

**Orthologs &
cross-organism**

**Metabolism
& chemistry**

**Sequence &
genome context**

3 AI RESEARCH ASSISTANT

Plans the analysis, writes and runs the code, and reports results with caveats

**Plans the
analysis**

**Runs reproducible
code**

**Reports results
+ caveats**

question

answer